

Introduction to C

A simple, clear walkthrough of what C is, how it works, and how a program runs — start to finish.

```
#include <stdio.h> → int main()
```



What is C?

C is a **general-purpose, procedural programming language** created by **Dennis Ritchie** in **1972** at Bell Labs, originally to build the UNIX operating system.

It's called a "**middle-level**" language — it gives you low-level control (like memory access) while still being readable like a high-level language.

Procedural

Compiled

Middle-level



Why Learn C First?

Builds logic

Teaches how computers actually process instructions

Foundation

C++, Java, Python all borrow from C's syntax

Close to hardware

Understand memory, pointers, and data directly

Used everywhere

OS kernels, embedded systems, competitive coding



Structure of a Program

Every C program follows the same basic skeleton:

```
// Documentation
#include <stdio.h>           // header

int main() {                 // entry point
```

```
int a = 5;           // declaration
printf("Hello!");   // statement
return 0;           // exit
}
```

Execution always starts at **main()** — no matter how big the program is.

⚡ How a Program Runs

1 Source Code

Your .c file, written in plain C syntax

2 Preprocessor

Expands things like #include before real compiling starts

3 Compiler

Translates code into object code (machine-level)

4 Linker

Connects your code with library functions (like printf)

5 Executable (.exe)

The final program — ready to run!



Tokens — The Building Blocks

Keywords

int, if, while — reserved words

Identifiers

Names you give — sum, total

Constants

Fixed values — 10, 3.14

Strings

"Hello World" in double quotes

Operators

+ - * / == &&

Symbols

{ } () ; ,



Your First Program

```
#include <stdio.h>
```

```
int main() {  
    printf("Hello, World!");  
    return 0;  
}
```

```
/* Output: Hello, World! */
```

printf() shows output. **scanf()** takes input — and always needs **&** before the variable name.

✓ Quick Recap

- C was made by **Dennis Ritchie** in **1972**
- Every program needs a **main()** function
- Code goes: **preprocess** → **compile** → **link** → **run**
- C is **case-sensitive**, statements end with **;**
- **printf** = output, **scanf** = input